Acoustic Leak Localization in Cast-Iron Water Mains

A. Fixture B. Fixture

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FICTIONAL TEST MATERIAL: this paper does not exist; it was written for the citation-audit fixture. No author, utility, venue, or measurement described here is real.

Abstract

Leaks on buried water mains are expensive to find because excavation is slow and a wrong dig is costly. We evaluate acoustic leak-noise correlators on a fictional municipal network and report how accurately they locate leaks on cast-iron mains up to 300 mm in diameter. Across 140 confirmed leaks, the correlators located 92% of them, with a median localization error of 1.5 m. We also compare the localization error on cast-iron, ductile-iron, and PVC mains, and find that the error grows as the pipe wall becomes less rigid.

1 Introduction

A leak-noise correlator places two acoustic sensors on fittings on either side of a suspected leak. The noise of escaping water travels along the pipe wall and through the water column to both sensors. The difference in arrival times, together with the distance between the sensors and the speed of sound in the pipe, gives the position of the leak between them. The method is old, but its accuracy depends strongly on the pipe material and diameter, and published field figures are scattered.

This paper reports a field evaluation on the fictional Fixture Valley network. Our question is narrow: how often, and how precisely, do correlators locate leaks on cast-iron mains, and how does that compare with other common pipe materials?

2 Method

2.1 Network and leak set

The evaluation covered cast-iron distribution mains with nominal diameters from 100 mm to 300 mm. A leak entered the study when a survey crew suspected

it and a later excavation confirmed it. In total 140 leaks on cast-iron mains were confirmed during the study period. A smaller comparison set of leaks on ductile-iron and PVC mains was collected in the same way.

2.2 Correlation procedure

For each suspected leak, the crew placed two sensors on hydrants or valves no more than 200 m apart. The correlator computed the cross-correlation of the two signals and reported the leak position. The crew then marked that position on the ground, and the excavation team recorded the true leak position after excavation. The localization error is the distance between the marked and the true position. A leak counts as *located* when the excavation at the marked position exposed it, which we define as an error of 3 m or less.

3 Results

3.1 Localization rate and error on cast iron

The acoustic correlators located 92% of the 140 confirmed leaks on cast-iron mains up to 300 mm in diameter. The median localization error on these mains was 1.5 m. Errors were smallest on the 100 mm and 150 mm mains and grew slightly with diameter, but the median stayed below 2 m in every diameter class. The leaks the correlators failed to locate were mostly at joints close to a branch, where the noise reached the sensors by more than one path.

3.2 Localization error by pipe material

Table 1 gives the median localization error by pipe material. Cast iron gave the smallest error, at 1.5 m. Ductile iron followed, at 2.1 m. PVC gave by far the largest error, at 4.8 m, because its flexible wall damps the leak noise and lowers the speed of sound along the pipe, so a small timing error turns into a large position error.

Table 1: Median leak-localization error by pipe material (Section 3.2).

Pipe material	Median localization error
Cast iron	1.5 m
Ductile iron	2.1 m
PVC	4.8 m

3.3 Practical consequence

On cast iron, a median error of 1.5 m means that a single standard excavation usually exposes the leak. On PVC, the crew should expect to widen the dig or to confirm the position with a second sensor spacing.

4 Limitations

This study measures leak localization only. It has clear limits on what it can support.

This study does not examine pressure management: network pressure was not changed during the study, and we draw no conclusion about how pressure affects leaks. This study does not examine bursts: every event in the data set was a background or reported leak, not a main break, and burst frequency was not recorded. This study does not examine water quality: no samples were taken and no water-quality parameter was measured. Finally, the comparison sets for ductile iron and PVC are smaller than the cast-iron set, so their medians are less certain.